

# Mohamed Elsayed Alshamy

## Computer Vision & Autonomous Systems Engineer

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### Objective

Computer Vision & Autonomous Systems Engineer with hands-on experience developing real-time computer vision and AI-powered robotic systems. Experienced in object detection, image processing, autonomous perception, and deep learning using Python, OpenCV, YOLO, and PyTorch. Hands-on experience with ROS2, Docker, FastAPI, Git/GitHub, Linux, databases, and embedded AI systems through autonomous vehicles, ROVs, and an AI-powered autonomous wheelchair.

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### Education Qualification

**Bachelor's Degree, Artificial Intelligence**

**Egyptian Russian University | ERU**

**Oct 2022 - Jun 2026**

**Activities:** ICPC - International Collegiate Programming Contest | 2024 & 2025

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### Selected Projects

**Autonomous Wheelchair – NEURONIX Intelligence Team | ERU**

**Oct 2025 – Jun 2026**

#### **Graduation Project**

- Developed an AI-powered autonomous wheelchair integrating **5 AI models**: object detection, face authentication, seatbelt detection, drowsiness detection, and floor-sign detection.
- Built a **3-camera system** for front/rear perception and user monitoring, integrated with **LiDAR** for navigation and obstacle perception.
- Designed the system around **NVIDIA Jetson Orin Nano, ESP32-WROOM, BTS7960, and ROS2**.
- Developed an onboard dashboard displaying **map, speed, distance, destination, time, user profile, and Manual/Autonomous modes**.
- Developed containerized AI and backend services using Docker and FastAPI, with Supabase for database and application data management.
- Received technical support from the **National Academy for Information Technology for Persons with Disabilities (NAID)**, Ministry of Communications and Information Technology.
- **Top 10 — ITC 2026, Robotics & Industry Track**; presented the project at the **Military Technical College Scientific Day**.

**Vision-Only Autonomous Driving — CARLA**

**Jan 2026 – Mar 2026**

- Developed a **vision-only autonomous driving system** using a single RGB camera, integrating **YOLOv8n** for multi-class object detection, camera-based distance estimation, threat assessment, and **Canny + Hough Transform** lane detection with temporal smoothing.
- Built a rule-based autonomous driving layer for **steering, acceleration, braking, traffic lights, stop signs, and obstacle avoidance**, achieving **15–17 FPS on CPU**, detecting 8 object classes with distance estimation up to 60m.

**Technologies:** Python, CARLA, YOLOv8, OpenCV, PyTorch.

## Autonomous Vehicle — Neutrino Team | ERU

Oct 2023 – Jun 2025

- Developed a **computer vision object detection system** on **Raspberry Pi 5** using **YOLOv8** and the **ZED 2 stereo vision camera** for real-time detection of traffic signs, stop signs, and road-related objects.
- Implemented **object tracking** for detected objects using the ZED 2 camera.
- Integrated the YOLOv8-based perception system with the vehicle's camera platform for real-time autonomous perception.
- **1st Place** — Valeo Egypt Autonomous Vehicle Competition 2024.

## Underwater ROV – Hydrotron Team | ERU

Oct 2022 – Jun 2025

- Developed an underwater **computer vision system** on **Raspberry Pi 4** using a front-facing camera and **YOLOv8** for real-time object detection, recognition and tracking.
- Developed **Arduino Nano** firmware for ROV control, including serial communication with the surface joystick and motor control for **six T200 thrusters**.
- Established **UART communication between Raspberry Pi 4 and Arduino Nano** to integrate the computer vision system with the embedded ROV control system.
- Integrated the camera, Raspberry Pi 4, Arduino Nano, joystick, and thruster control into a unified underwater ROV system.
- Achieved **8th Place — MATE ROV International Competition 2023**, **6th Place — MATE ROV International Competition 2024**, and **4th Place — MATE ROV International Competition 2025**.

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## Internships and Certifications

- Computer Vision – Autonomous Systems, NAID | Apr 2026 – May 2026
- Deep Learning – Computer Vision, NTI | Jul 2025 – Sep 2025
- Embedded Systems, ITI | Jul 2025 – Aug 2025
- Artificial Intelligence Level 3, Orange Digital Center | Sep 2024 – Oct 2024
- HCIA-AI, Huawei ICT Academy | Jul 2024 – Sep 2024

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## Technical Skills

- **Computer Vision:** Object Detection, Object Tracking, Image Processing, Lane Detection, Face Recognition, Scene Understanding, OpenCV, YOLO
- **AI & Deep Learning:** PyTorch, TensorFlow, Keras, CNNs
- **Robotics & Autonomous Systems:** ROS2, ROS, Autonomous Navigation, CARLA, Motion Planning, Real-Time AI
- **Embedded & Hardware:** NVIDIA Jetson, Raspberry Pi, ESP32, Arduino, LiDAR, CUDA
- **Programming:** Python, C, C++, Java, MATLAB
- **Backend & Data:** FastAPI, REST APIs, Supabase, Database Integration
- **Development Tools:** Git, GitHub, Kafka, Docker, Jupyter, Google Colab, VS Code
- **Operating Systems:** Linux, Windows

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## Languages

- Arabic: Native
- English: Fluent
- French: Intermediate

